

XRPL Validator — Daily Engineering Report • 2026-09-28

Generated 2026-09-29 00:21 EDT. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the live native-engine shadow, the reference rippled node, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **2 commits** (xrpl-validator 2)
- **1 engine finding fixed** (F406)
- Live shadow: 6,333/6,333 ledgers fully matched, 842,788 transactions, 0 engine receipts in the current soak
- Amendment **PermissionDelegationV1_1** held majority (since 2026-09-24 21:25 UTC); activation due ~2026-10-08 21:25 UTC
- Amendment **fixBatchV1_2** held majority (since 2026-09-25 14:12 UTC); activation due ~2026-10-09 14:12 UTC
- Amendment **BatchV1_1** held majority (since 2026-09-25 14:46 UTC); activation due ~2026-10-09 14:46 UTC

Live validator and native engine

Snapshot at generation time.

Measure	Value
Ledgers compared (current run)	6,333
Ledgers fully matched	6,333
Transactions applied natively	842,788
Result mismatches	0
Byte mismatches	0
Missing / extra keys	0 / 0
Batch inner-result mismatches	0
Engine receipts (current soak)	0
Live engine build running since	2026-09-28 17:30 EDT
Reference rippled	3.4.1, full, validated #107,307,310

Engine findings fixed

- **fix(xrpl-ledger): F406 — an AMM equal withdrawal pays the sides in the pool object's order** (0145757, xrpl-validator) Mainnet #107299319 1E5C6C1A9257: a tLPToken withdrawal from the XRP/BCFT pool by an account holding 9.381031 XRP at OwnerCount 41 with no BCFT trust line. The transaction names the pool as Asset = BCFT, Asset2 = XRP; the AMM object stores it as Asset = XRP, Asset2 = BCFT. With no Amount or Amount2, AMMWithdraw::applyGuts asks ammHolds for the pool with no assets, which answers in the object's order, so rippled pays ...

Engineering changes

- **fix(xrpl-ffi): link the conan libraries in a fixed order; skip gRPC** (21fdd93 , xrpl-validator)
collect_conan_libs() walked the conan generators directory with fs::read_dir, whose order is the filesystem's (hash order on ext4), so the static-library link order, and with it which archive member supplies a duplicated C++ symbol, differed between hosts. On the build box the differential_probe link failed: a rocksdb object needed a std::string instantiation, the first supplier in that host's order was a gRPC ...

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Soak #33: one engine receipt, fixed.** Soak #33 of build ae84a26 ran from 08:28 EDT on 2026-09-27 to 08:32 EDT today and recorded one engine receipt, F405 below. Since that build's deployment at 08:01 EDT on 2026-09-27 the shadow compared 22,690 ledgers and the state-hash check matched 22,701, with no state-hash mismatch: the receipt is a difference in the native engine's own result, found by its comparison against libxrpl, not in the validator's state. The validator stayed on its primary reference node throughout, with no failovers.
- **Finding F405 (AMM withdrawal by the token's issuer).** At ledger 107,278,773 an account withdrew all of its share of an AMM pool of XRP and a token the account itself issues, while holding 1.41 XRP and owning two ledger objects. Before paying out a token, rippled checks that a withdrawer with no trust line for it could afford the reserve for a new one. The check asks only whether that trust line exists, and an issuer never has a trust line to itself, so it applies to the issuer too: a third object needs $1 + 0.2 \times 3 = 1.6$ XRP, and even with the pool's whole 0.1 XRP the account had at most 1.50999 XRP, so mainnet refused the withdrawal (tecINSUFFICIENT_RESERVE, with the fee charged). The engine exempted issuers from the check, an assumption made when the check was added on 2026-09-04 that no mainnet transaction had tested until this one, and paid out. The exemption is removed and the mainnet transaction is pinned as a regression vector. Fixed in 51b0cbe on 2026-09-27 and deployed today (below).
- **Deployment.** Build 21fdd93 replaced the live build at 10:09 EDT, after its gate (1,692 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.1 seconds and resumed two ledgers behind the network. The health check matched 401 ledgers with no mismatches, and soak #34 began at 10:35 EDT (seed ledger 107,294,149). Besides F405 it fixes how the build links libxrpl's static libraries: they were linked in the order the filesystem listed them, which differs between machines, so the same link succeeded on one machine and failed on another. They are now linked in a fixed order, and gRPC, which nothing uses, is no longer linked.
- **Soak #34: one engine receipt, fixed.** Soak #34 of build 21fdd93 began at 10:35 EDT and ended about seven hours later with the deployment below. It recorded one engine receipt, F406. Every check during it found no state-hash mismatch (the last at 15:17 EDT); the receipt is a difference in the native engine's own result, found by its comparison against libxrpl, not in the validator's state.
- **Finding F406 (AMM withdrawal: which side is paid first).** At ledger 107,299,319 an account holding 9.381031 XRP and owning 41 ledger objects withdrew its share of an XRP/token pool by redeeming LP tokens, with no trust line for the token. The transaction named the token first and XRP second, but the pool's ledger object stores XRP first, and when a withdrawal names no amounts rippled pays the two sides in the object's order. So the XRP arrived first and counted toward the 9.4 XRP reserve the new trust line needed, and mainnet accepted the

withdrawal. The engine paid the sides in the transaction's order, checked the token side against 9.381031 XRP and refused it (tecINSUFFICIENT_RESERVE). It now pays proportional withdrawals in the pool object's order, as its deposit code already did. The mainnet transaction is pinned as a regression vector. Fixed in 0145757.

- **Deployment.** Build 0145757 replaced the live build at 17:31 EDT, after its gate (1,693 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.6 seconds and resumed two ledgers behind the network. The health check matched 418 ledgers with no mismatches, and soak #35 began at 17:58 EDT (seed ledger 107,300,970). It was the first deployment run from the operations team's version-controlled deployment scripts.
- **fixBatchV1_2.** Its source is still unpublished (last checked 21:17 EDT on 2026-09-27); it activates on mainnet about 2026-10-09 14:12 UTC.

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Issuer-set TickSize for MPT order books** (dex) Add an optional sfTickSize (UINT8, 3–15) to MPTokenIssuance. It is set at MPTokenIssuanceCreate and can be changed later through MPTokenIssuanceSet on
- **Permissionless batch sweep of expired objects** (fees) Add ttEXPIRED_SWEEP, which carries sfSweepIDs, an STVector256 of at most 16 entries. Any account may submit it. Each keylet must resolve to an object
- **Oracle-banded execution guard for Payment/OfferCreate** (dex) Add an optional sfOracleGuard inner object to Payment and OfferCreate. It holds sfOracleID (the keylet of a g9 AggregateOracle or a single PriceOracle

Appendix — commits

Time	Repo	Commit	Subject
09:10	xrpl-validator	21fdd93	fix(xrpl-ffi): link the conan libraries in a fixed order; skip gRPC
16:04	xrpl-validator	0145757	fix(xrpl-ledger): F406 — an AMM equal withdrawal pays the sides in the pool object's order